Constructor University

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Lab Experiment 1: Usage of Oscilloscope

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1.Introduction

The oscilloscope is one of the most essential instruments in electrical engineering, because it allows direct visualization of electrical signals in the time domain

The objective of this experiment was to get familiarized with one of the most essential tools in Electrical Engineering. In this experiment, the basic operating principles of the oscilloscope were studied, including the use of vertical and horizontal controls, triggering, coupling modes, cursor measurements, automatic measurements, and probe compensation.

An oscilloscope allows visualization of voltage signals as a function of time: Voltage (amplitude, peak-to-peak), Time (period, frequency) and phase relationship.

The experiment focuses on:

1. Vertical control (voltage measurement)
2. Horizontal control (time and frequency)
3. Trigger system
4. Probe usage and compensation

Understanding this properties gives us capability to analyze AC and DC Circuits.

Understanding these functions enables accurate signal analysis in both AC and DC circuits.

The vertical axis of the display represents voltage, while the horizontal axis represents time. Voltage measurements on an oscilloscope are performed using the relation:

<u>V = (number of division) (Volts/div).</u>

Similarly, the time period T of a signal can be determined from the horizontal scale, and the frequency is given by:

<u>f = 1/ T</u>

This experiment focuses on understanding these fundamental principles and applying them to measure voltage, time, and frequency characteristics of electrical signals. This experiment was structured around four parts: vertical control, horizontal control, the trigger system, and probe usage.

2.Execution

2.1 Experiment Setup

Workbench No.6

Instruments Used:

- Oscilloscope
- Signal generator
- Passive probe
- Power supply
- Breadboard
- Connecting wires

The passive probe was connected to Channel 1 of the oscilloscope. The signal generator output was connected with a BNC cable, and the probe ground clip was connected to circuit ground.

The initial generator settings were:

- Function = Sine Wave
- Frequency = 60 Hz
- Amplitude = 0.6 Vpp
- Offset = 0 V

Part 1: Vertical Control (Voltage Measurement)

Procedure

The objective of this part was to study the effect of vertical and horizontal controls on waveform. A sine wave signal was generated and displayed on the oscilloscope. Voltage was measured using three methods:

1. Graticule (grid counting)
2. Cursor method
3. Measure function

Results

Table 1: Voltage Measurement using Different Methods

Method	Range	V_{min}	V_{max}	V_{PP}	Estimated Resolution V/[div]
Graticule division	1	2.2	2.8	0.6	0.1

cursor	1	2.24	2.8	0.56	0.1
measure function	1	2.24	2.8	0.56	0.1

Observation

- Increasing sensitivity (lower V/div) improves accuracy.
- Cursor and measure functions give more precise results than manual counting.

Switching Input Coupling

Investigate how different input coupling modes (AC, DC) affect the displayed signal and enable measurement of small AC components in the presence of a DC offset.

Measure	Minimum	Maximum	Mean	Peak-Peak
Value	-284mV	272mV	-3.95mV	556mV

Observation

The cursor and automatic measurement values were nearly identical, while the graticule method showed larger reading uncertainty.

The period of the waveform was measured using the horizontal control and found to be:

T = 17.2 ms

Part 2: Usage of the Horizontal Control

Objective

The objective of this part is to measure **time-related properties** of signals using the oscilloscope, including period, frequency, and time differences, by utilizing the horizontal control and cursor functions.

Experimental Setup

Workbench No 6

Used Instruments

1. Oscilloscope
2. Signal generator

3. Passive probe
4. Connecting cables

The oscilloscope was connected to the signal generator output using Channel 1 as the reference signal. A rectangular waveform and a sinusoidal waveform were displayed simultaneously on the screen in order to observe period, frequency, and time delay.

Execution

The displayed waveform consisted of three components:

- A rectangular signal
- A sinusoidal signal
- A time delay between them

Results

Table 2: Rectangular Wave Measurements

Rectangle	Period	Frequency	Amplitude
	17.2 ms	58.14Hz	5.2

Table 3: Sine Wave Measurements

signal	Frequen-cy	Amplitude max(V)	Amplitude min(V)	Vpp (V)
Sine Wave	238 Hz	2.84	2.28	0.56

Table 4: Time delay

Beginning of the signal (from starting rectangle to sine wave)	9.6 ms

Observation

We observed that expanding waveform across screen improves accuracy, and using cursors are necessary for complex signals.

Part 3: Trigger Section

Objective

The objective of this part is to understand the function of the oscilloscope trigger system and how trigger settings affect signal stability and display. Trigger level was changing manually while observing waveform behavior

Experimental Setup

Workbench No.6

Used Instruments

- Oscilloscope
- Signal generator
- Passive probe
- Connecting cables

The oscilloscope is first initialized using the Default Setup and Autoset functions. The signal is adjusted such that it occupies a large portion of the display to improve measurement accuracy. The trigger settings are configured with Type = Edge, Source = CH1, and Mode = Auto.

Execution and Results:

The trigger level was varied from approximately −1 V to +6 V. When the trigger level was within the signal amplitude range, the waveform appeared stable. Outside this range, the signal became unstable or disappeared.

Around 2.5 V, the waveform showed instability and horizontal movement.

When changing from Auto to Normal mode, the waveform was only displayed when a valid trigger condition was present.

Observation

-1 ~ 0.16 is unstable state. / 0.2 ~ 2.5 is stable. And 2.5 is unstable state and a lot of moving horizontally. 2.6 ~ 5 stable . 5.5 ~ 6 is unstable. Difference between rising and falling setup is unstable point.
Other important trigger controls : disappear until 480mV

Part 4 : Using the Probe

Setup

The aim of this part is to understand how the probe affects the signal and why proper compensation is important. A passive probe is used and connected to the oscilloscope.

Execution and Results

A correctly compensated probe shows a clean square wave. If the probe is under-compensated, the edges appear rounded. The adjustment is tuned until the waveform becomes a proper square signal.

Three compensation conditions were observed:

- Undercompensated probe → rounded waveform edges
- Overcompensated probe → overshoot at transitions
- Correctly compensated probe → ideal square wave

Proper compensation makes sure that probe capacitance matches oscilloscope input capacitance and prevents waveform distortion.

3. Evaluation

3.1 Comparison of Measurement Methods

The measured values of voltage and time were obtained using different methods, including the graticule, cursor, and automatic measurement functions. We found out that the accuracy improves when the signal occupies a larger portion of the screen.

Graticule method was least accurate because it depends on human reading, and estimation.

Cursor method improved precision because of digital reference points.

Measure function was most accurate (calculated with oscilloscope), which was done by calculating internally

The measured period was: T = 17.2 ms

Therefore: $f = \frac{1}{17.2} = 58.15Hz$

This value is close to the generator setting of 60 Hz, confirming measurement accuracy.

Small difference was mainly caused by reading uncertainty.

3.2 AC/DC Coupling

DC coupling displays the complete waveform, including both the AC signal and the DC offset component.

In contrast, AC coupling blocks the DC component and displays only the varying AC part of the signal. This allows small AC variations to be observed more clearly when they are superimposed on a large DC voltage.

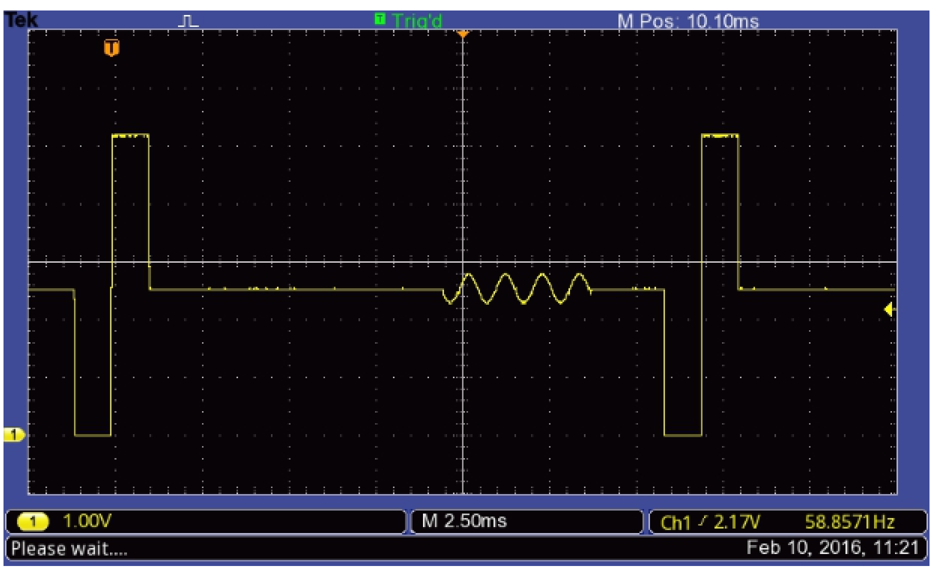
Ground coupling disconnects the input signal and displays the zero-voltage reference line, which is useful for establishing a correct baseline during measurements.

During the experiment, it was observed that switching from DC coupling to AC coupling shifted the waveform toward the zero reference line, confirming the removal of the DC offset.

The experimental results matched the theoretical behavior of the three coupling modes and showed the importance of selecting the correct coupling setting depending on the type of signal being measured.

3.3 Evaluation of Part 2: Horizontal Control

The horizontal control allowed accurate measurement of time intervals and frequencies using the cursor function. Compared to the automatic measurement tool, the cursor method was more reliable for complex signals like containing multiple components. The measured time delay between the rectangular waveform and the sine wave was 9.6 ms, demonstrating the usefulness of the oscilloscope for timing analysis and phase comparison between signals. The experimental results confirmed that the horizontal control is essential for accurate frequency and time-domain measurements.



3.4 Evaluation of Part 3: Trigger Section

The trigger system was essential for stabilizing the waveform. A stable signal was obtained only when the trigger level was within the signal amplitude range. Outside this range, the signal became unstable or disappeared.

In Normal trigger mode, the oscilloscope required a valid trigger signal to display the waveform, while in Auto mode, the oscilloscope forced a sweep without valid trigger condition.

This demonstrated that proper trigger adjustment is necessary for accurate and stable signal observation.

3.5 Evaluation of Part 4: Using the Probe

Incorrect probe compensation caused distortion in the waveform, such as rounded edges or overshoot. After proper adjustment, a clean square wave was obtained, confirming correct compensation. 3

This occurs because the probe capacitance must match the oscilloscope input capacitance. Proper compensation ensures accurate waveform reproduction and prevents measurement errors.

4. Conclusion

In conclusion, the experiment successfully demonstrated the main functions of the oscilloscope, including voltage, time, and frequency measurements. The results showed that measurement accuracy improves when the signal is properly scaled and used appropriate tools, such as cursors or automatic measurements.

The experimental observations were matching with theoretical expectations. For example, the relationship between period and frequency was confirmed, and the behavior of AC coupling, triggering, and probe compensation matched the expectation.

There were some small differences and errors which can be explained mainly due to limited resolution of oscilloscope, mispositioning cursors, and noise in signals. Additionally, improper trigger settings or probe compensation could temporarily distort the waveform. These highlights the importance of choosing correct instrument and functions.

Overall, the experiment provided a clear understanding of oscilloscope operation and the factors that influence measurement accuracy.

5.Reference

Uwe, P. (2024, 11 21). Retrieved from Uwe Pagel - Constructor University Natural Sciences Laboratory: URL